Automated Refueling Cislunar Outpost (ARCO)

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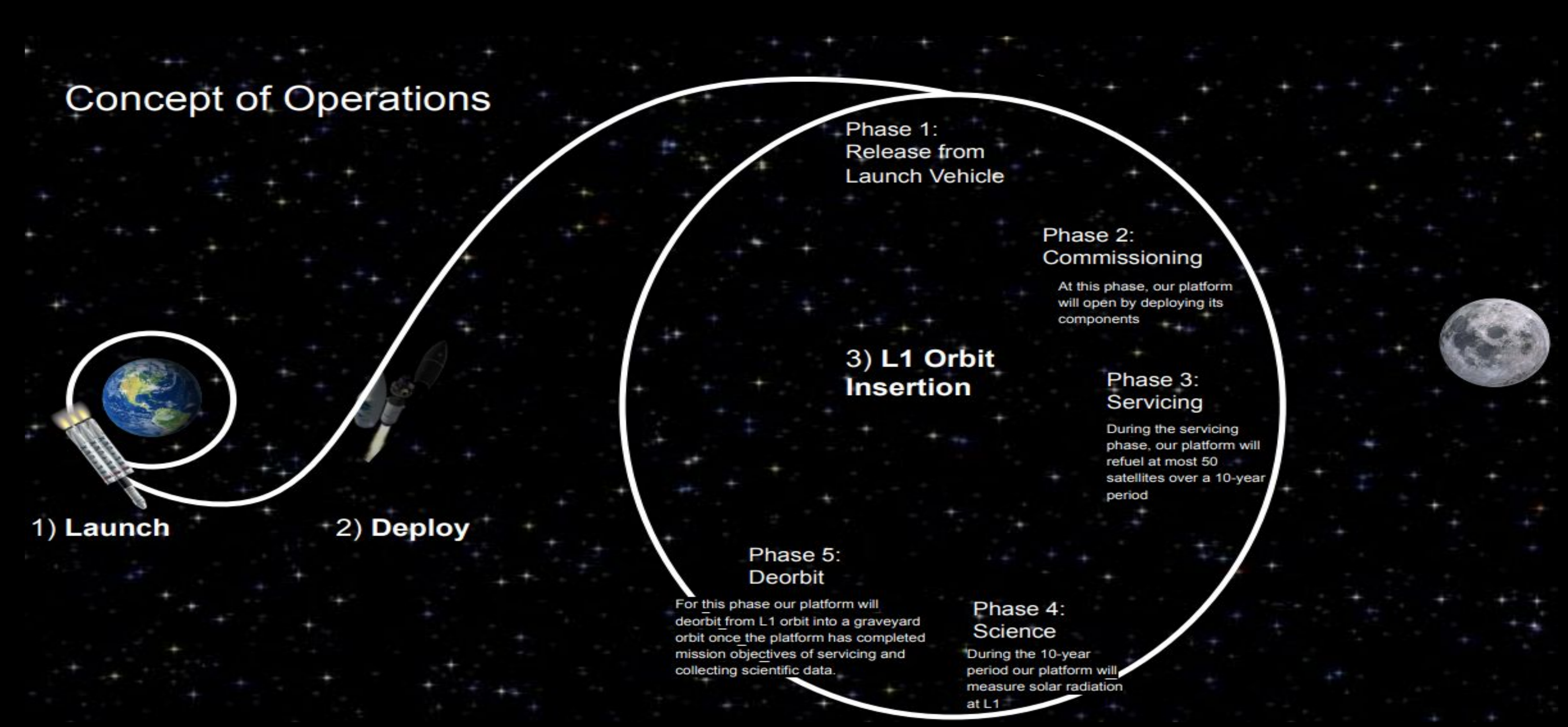
<u>Mission Statement:</u>

Automated Refueling Cislunar Outpost (ARCO) will be launched into a LEO parking orbit by the Falcon Heavy launch vehicle. From this parking orbit the Falcon Heavy will perform a trans-L1 injection, sending ARCO on a trajectory to the Earth-Moon L1 point. Upon reaching L1, Phase 1 will begin and the L1 orbit insertion maneuver will occur. After the insertion, ARCO will be released from the launch vehicle. In Phase 2, commissioning occurs. In this phase, ARCO will commence deployment of its components and perform all necessary tests to ensure everything is in working condition after launch. At Phase 3, servicing begins and our platform will refuel approximately 50 mid-size spacecraft with 225 kg of Hydrazine each for a total of 11,250 kg of Hydrazine over the 10-year mission life. ARCO will be continually pointing its high gain antenna at the Earth, and solar panels will be articulated to provide constant power to the solar panels. The orbit will have to be maintained by the SCAT thrusters with a low delta-v maneuver approximately every 21 days. Reaction wheel momentum will have to be offloaded by the MRE thrusters with low delta-v burns approximately every 23 days. During Phase 4 (which will run concurrently with Phase 3) ARCO will perform radiation assessments of the L1 environment. At the end of mission life, phase 5 will begin, at which point our platform will cease orbit maintenance, falling into a high Earth orbit.



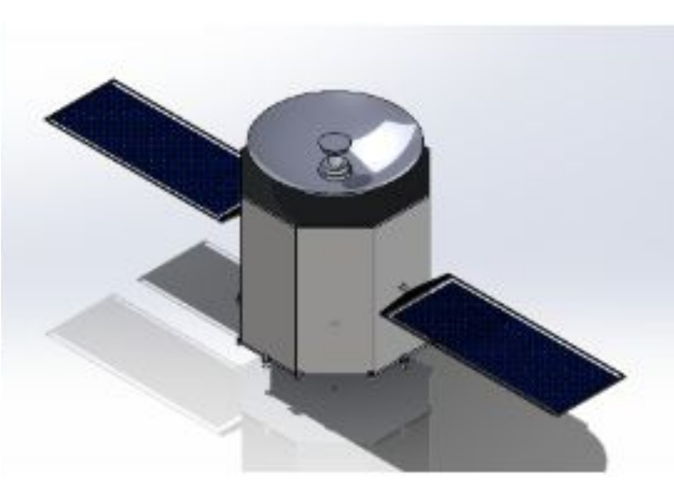
<u>Avionics:</u> Oversees science, communication, and power. The power usage is 1503.59 W per day assuming all components are in operation. Will be using the Radiation Assessment Detector for collection of radiation. Will use the Oxford Space System Antenna for communication. For power will be using Roll-out Solar Arrays due to their light mass.

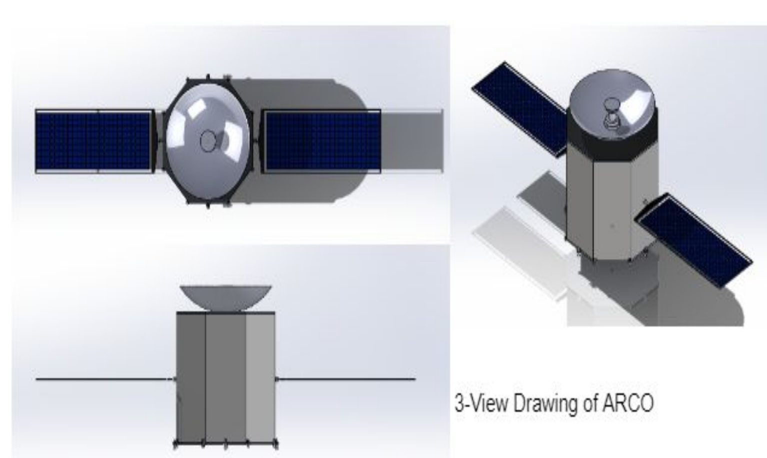
<u>Structures:</u> Enables the spacecraft to survive six times the force of gravity during launch as well as protect vital electronics. Focuses on the implementation of a 50 mm thick shield for radiation/debris protection as well as an inner frame used to withstand the force of launch. Also involves the structural design of the solar arrays.

<u>Propulsion:</u> Responsible for station keeping, momentum unloading and attitude control with Northrop Grumman MRE-1.0 and SCAT TR-500

<u>Guidance, Navigation, and Control:</u> Allows the spacecraft's antenna to stay pointed towards the Earth allowing for continuous communication. The GNC subsystem also uses star trackers with an accuracy of 5 arcseconds to determine our platform's exact position in space.

<u>Docking:</u> For docking the RAFTI system will be used to refuel and dock with other spacecraft. With the use of 3 fiducials, ARCO will communicate with customer spacecraft to do a co-operative dock. Once docked, the RAFTI system will be used to transfer hydrazine fuel. The fuel transfer rate of the RAFTI system is 4 L/min. Therefore in just under an hour, ARCO will be able to refuel a customer spacecraft with 225 kg of hydrazine propellant.





3-View Drawing of ARCO

Cost Table:

Subsystem	Cost ($)	% Overall Cost
Docking	54,000	7.21%
Avionics	14,844,000	18.46%
Propulsion	48,000,000	70.02%
GNC	1,558,000	2.37%
Structures	1,278,000	1.94%
Total:	**65,734,000**	**100%**

Mass Table:

Subsystem	Mass (kg)	Overall Mass (%)
Docking	0.550	0.3%
Avionics	252.95	1.5%
Propulsion	11,542.690	67.05%
GNC	7.185	4.33%
Structures	4,779.939	26.82%
Total:	**16,583.314**	**100%**

Andy	Team lead/Tank thickness	Tasked team members and kept the mission timeline. Used excel & material data to verify tank could handle operating pressure.
Andrea	Solar panel actuator rotation	Used Simulink to verify the actuators will produce enough torque to rotate the panels to their new required location.
Juan	Reaction wheel sizing	Used MATLAB to calculate the amount of torque required to maneuver the spacecraft using reaction wheels
Ernest	Launch axial load analysis	Used Solidworks Finite Element Analysis on individual corners to verify that structure does not observe permanent deformation.
Elizabeth	Energy produced by Solar Panels	Calculated power used by components. While also calculating the energy needed to sustain our spacecraft during an orbit.
Summer	Stress Analysis of propulsion	SolidWorks was used to model the thrusters and analyzed the stress on the model when propulsion force is on
Angelica	Solar panel launch load analysis	Ran stress test on Solidworks using 3,000 N of force at the solar panel center of mass
Alex	Vibration analysis	Solidworks structural analysis simulating launch vibration loads.
Samuel	Tank thermal analysis	Solidworks thermal analysis to verify hydrazine can be kept at operating temperature 2-113 degrees C.

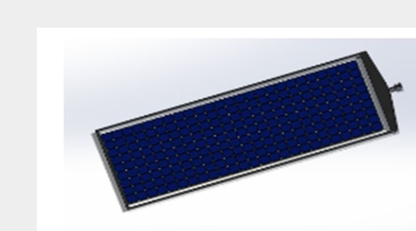
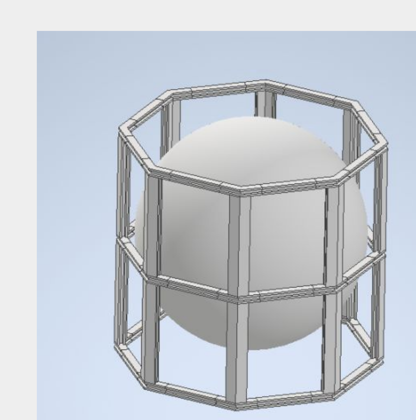
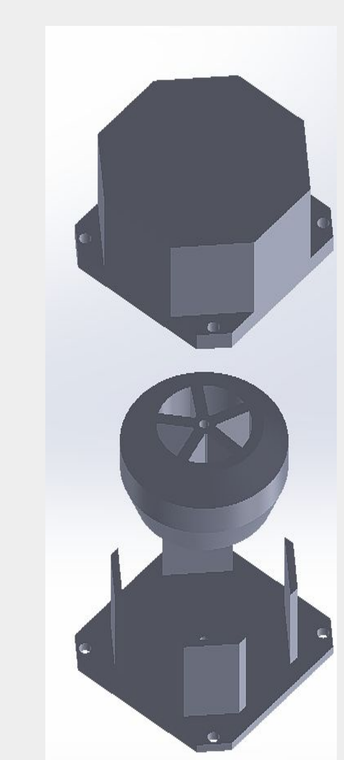
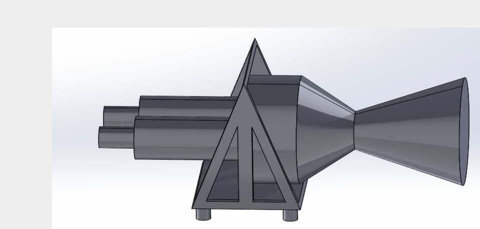
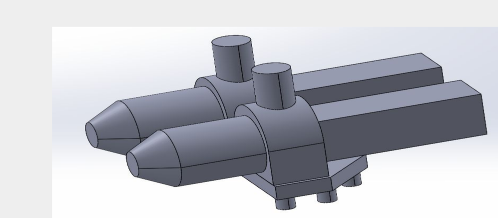


Figure 7.6: Model of Solar Array









- Budget: $65,739,000
- Mass: 16,583.314 kg
- Power: 1.504 kW
- Number of Spacecraft Refueled: 50
- Life Expectancy: 10 years